

NEXONIX INTERNSHIP

Assignment 02 of 04 — Drone Aerodynamics, Propulsion & Electronics

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Q1. Flight Stability Analysis

Question: A quadcopter is unable to maintain a stable hover and continuously drifts during flight. Identify any four possible aerodynamic or mechanical reasons and explain how each issue affects flight performance.

1. Unequal Thrust Distribution

If the four motors do not produce the same amount of thrust, the drone may tilt slightly toward the weaker side. This makes the thrust direction uneven and a part of the thrust acts sideways. As a result, the drone slowly drifts instead of staying at the same point.

2. Centre of Gravity (CG) Offset

If the battery, camera or payload is placed away from the centre, the centre of gravity also moves. The flight controller then has to keep correcting the motor speeds. If the imbalance is large, the drone can keep moving toward the heavier side and the hover becomes less stable.

3. Damaged or Mismatched Propellers

A bent, chipped or wrongly matched propeller will not produce the same lift as the other propellers. It can also create extra vibration. This makes the drone wobble and can affect the sensor readings, leading to irregular corrections by the flight controller.

4. Wind or Turbulence

A crosswind produces a sideways force on the drone and pushes it away from its position. The controller tries to compensate by changing the motor speeds, but strong or changing wind can still cause noticeable drift, especially when the drone has limited spare thrust.

Conclusion: A stable hover needs balanced motor thrust, a properly placed centre of gravity, good propellers and enough thrust to handle outside disturbances. If any of these are not correct, the drone may drift or wobble.

Q2. Electronics & Sensor Analysis

Question: During an autonomous mission, the drone loses GPS signal and starts showing unstable flight behaviour. Explain the role of the following components in maintaining stable flight, and what may happen if any two fail during flight.

Component	Role in Stable Flight
Flight Controller	It is the main control unit of the drone. It receives sensor data, compares the actual motion with the required motion, and adjusts the motors to keep the drone stable.
IMU (Accelerometer & Gyroscope)	The IMU measures acceleration and angular movement. The flight controller uses this information to detect tilting, rotation and movement and make quick corrections.
GPS Module	GPS provides the drone's position, speed and direction. It is especially useful for position hold, navigation and autonomous missions.
Magnetometer / Compass	It senses the Earth's magnetic field and helps the drone know its heading. This is useful for navigation and maintaining the correct direction.
Barometer	It measures air pressure and helps estimate altitude. The flight controller uses it to maintain a steady height, especially during automatic flight.

What can happen if two components fail?

Example 1 — GPS failure: If GPS fails during autonomous flight, the drone may not be able to hold its position or follow the planned route. It can start drifting with wind unless another navigation method is available.

Example 2 — IMU failure: If the IMU gives incorrect data or stops working, the controller may not know the drone's actual tilt and rotation correctly. This can cause wobbling, wrong motor corrections and, in a serious case, loss of stable flight.

Conclusion: Stable flight depends on several sensors working together. The flight controller combines their information to control the motors. A failure in an important sensor can reduce stability, accuracy and safety during flight.